

29	Answer: D
	Activity $A = A_0 \exp(-\lambda t)$, where A_0 is the activity at $t = 0$. The decay constant is given by $\lambda = \frac{\ln 2}{t_{1/2}}$. A larger half-life yields a smaller λ. Hence, $\lambda_C > \lambda_N$. Taking natural log, $\ln A = -\lambda t + \ln A_0$. The graph of $\ln A$ vs t <u>for each nuclide</u> is a straight line with negative gradient $(-\lambda)$. Initially, because of its short half-life, the activity of nuclide C dominates. So the initial part of the graph should be very close to that due to nuclide C alone, with gradient $-\lambda_C$. Later, nuclide C have mostly decayed, so nuclide N starts to dominate. The later part of the graph should be close to that due to nuclide N alone, with gradient $-\lambda_N$. Remembering that $\lambda_C > \lambda_N$, the graph should have less negative gradient (a gentler downward slope) in the later part than in the earlier part.

30	Answer: B
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